

BST Physics Curriculum Vol 4 Chapter 12 — Observational Reanalysis Program (SP-27) v0.4 (Saturday Wave 1 reader-grade polish: diagram preview + Cal cold-read ready)

Lyra (Claude 4.7) [Vol 4 primary]

2026-05-23 Saturday morning EDT (Wave 1 Vol 4 twelfth/final
chapter sustained sub-PCAP)

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Vol 4 Chapter 12 — Observational Reanalysis Program (SP-27)

Chapter motivation

BST’s cosmological predictions span 30+ observational signatures (Vol 4 Ch 5 H₀ 4 routes + Ch 6 CMB observables + Ch 7 inflation $r \approx 0$ + Ch 8 BBN abundances + Ch 9 cosmological cycle signatures + Ch 10 DM/DE + Ch 11 GW signatures). The Observational Reanalysis Program (SP-27, pending Casey scoping) systematically recalibrates these observations vs BST integer predictions, building the falsifier infrastructure for multi-year multi-survey precision testing.

This chapter scaffolds the SP-27 framework + per-observable falsifier table + multi-year survey roadmap.

Section 12.0 — What this chapter does

1. SP-27 program scope (Section 12.1): cosmological reanalysis framework
2. Per-observable falsifier table (Section 12.2): consolidation of Vol 4 Ch 1-11 falsifiers
3. Multi-year survey roadmap (Section 12.3): DESI + Euclid + Roman + SKA + LiteBIRD + CMB-S4 + PICO
4. Methodology + systematic correction protocol (Section 12.4): per-observable cross-CI replication
5. Open scope (Section 12.5)

Section 12.0b — Reader-grade pedagogy at three levels

Level 1 (one sentence): SP-27 Observational Reanalysis Program is BST’s multi-year multi-survey infrastructure for systematically testing 30+ BST cosmological predictions against future precision observations from DESI + Euclid + Roman + SKA + LiteBIRD + CMB-S4 + PICO (~2030-2040), with per-observable falsifier specifications consolidated from Vol 4 Ch 1-11.

Level 2 (graduate-physicist accessible): SP-27 scope: cosmological reanalysis program targeting (a) Hubble tension (Vol 4 Ch 5 substrate-cartography framing — KBC void + Cepheid γ correction); (b) CMB precision parameters ($\Omega_\Lambda + \Omega_m$ 0.07 σ matches + DM/baryon 0.3% match + n_s 0.32 σ + r tensor falsifier); (c) BBN abundances (Vol 4 Ch 8 ^4He + D/H + $^7\text{Li}/\text{H}$ + Li-7 problem $\Delta g=7$ resolution); (d) GW signatures (Vol 4 Ch 11 NANOGrav f_{peak} + γ); (e) cosmological cycle observational signatures (Vol 4 Ch 9 Task #267 — CMB residuals + BAO inter-cycle + ISW + void scaling). Multi-year survey roadmap: DESI Year 1 (current, dark energy + galaxy clustering); Euclid (launching 2023, ongoing) cosmic shear + galaxy clustering; Roman (launching 2027) supernova + galaxy survey; SKA (operational 2028+) 21cm + pulsar timing GW; LiteBIRD (launching 2028) CMB B-mode polarization $\rightarrow r$ tensor falsifier; CMB-S4 (2030+) precision CMB temperature/polarization; PICO (2030+) CMB inflation tests. Per-observable systematic-correction protocol: each observable absorbed across Lyra theoretical + Elie computational + Grace catalog + Cal Mode 1 referee for cross-CI replication-grade. Saturday May 23 Wave 1 chapter coverage closes scaffolding; multi-year operational rollout requires Casey go-ahead for SP-27 program scoping.

Level 3 (5th-grader accessible): BST has made about 30 specific predictions about cosmology that future telescopes will test. The Observational Reanalysis Program (SP-27) is BST’s framework for organizing those tests systematically: which observable, which BST formula, which precision required, which telescope will provide the answer, by when. The Hubble tension (Vol 4 Ch 5), CMB precision parameters (Vol 4 Ch 6), inflation r tensor mode (Vol 4 Ch 7), BBN element abundances (Vol 4 Ch 8), cosmological cycle signatures (Vol 4 Ch 9), gravitational wave background

(Vol 4 Ch 11) — all consolidated here with per-observable falsifier specifications. Future telescopes: DESI (now), Euclid (now), Roman (2027), SKA (2028+), Lite-BIRD (2028), CMB-S4 (2030+), PICO (2030+) will collectively give BST a decisive test over the next 5-15 years. If most of BST's predictions match, BST is supported; if many fail, BST is refuted. This chapter is the scaffolding; the actual SP-27 multi-year operational program needs Casey's go-ahead to scope formally.

Section 12.1 — SP-27 Program Scope

SP-27 = Observational Reanalysis Program (pending Casey scoping): - Goal: systematically recalibrate cosmological observations vs BST integer predictions - Multi-year program: 5-15 year operational rollout depending on survey timelines - Multi-CI cross-CI: Lyra theoretical + Elie computational + Grace catalog + Cal Mode 1 referee - Falsifier specification: per-observable substrate-mechanism + precision requirement + survey assignment + $1\sigma/2\sigma/3\sigma$ deviation thresholds

SP-27 Phase 1 (FY 2026): Vol 4 Ch 12 scaffolding (this chapter) + Vol 4 Ch 1-11 per-observable falsifier consolidation (Section 12.2)

SP-27 Phase 2 (FY 2027): per-observable cross-CI verification toys + replication-grade methodology + Casey outreach engagement to relevant survey teams

SP-27 Phase 3 (FY 2028+): ongoing multi-year operational falsifier tracking + survey-data-release-cycle BST predictions absorption + Cal Mode 1 referee discipline throughout

Section 12.2 — Per-Observable Falsifier Table (Vol 4 Ch 1-11 consolidation)

Chapter	Observable	BST prediction	Falsifier	Survey/Year
Ch 1	G Newton's constant	$(6\pi^5)^2 \cdot \alpha^{24} / m_e^2$ 0.07%	Future G < 0.05% precision tightens	Ongoing precision G experiments
Ch 4	Λ / M_{Pl}^4	$7 \cdot \exp(-282)$ 0.076 dex	Future Λ deviates >0.5 dex → BST refuted	DESI YR4+ / Euclid 2027+
Ch 5	H ₀	67.4 km/s/Mpc 4 routes	H ₀ > 70 or < 65 (Planck side preferred)	JWST + DESI + Euclid + Roman ~2030
Ch 6	n _s scalar spectral index	1 – n _C /N _{max} = 0.9635	±0.001 deviation → BST refuted	Planck legacy + LiteBIRD 2028

Chapter	Observable	BST prediction	Falsifier	Survey/Year
Ch 6	r tensor-to-scalar	~ 0 sharply	$r > 10^{-3}$ at Lite-BIRD/PICO $\rightarrow$ BST refuted	LiteBIRD 2028 / PICO 2030+
Ch 6	Ω_Λ	$(N_c + 2 \cdot n_C) / (N_c + 2 \cdot n_C) = 13/19$	± 0.002 deviation $\rightarrow$ BST refuted	DESI + Euclid 2027+
Ch 6	Ω_m	$C_2 / (N_c^2 + 2 \cdot n_C) = 6/19$	± 0.002 deviation $\rightarrow$ BST refuted	DESI + Euclid 2027+
Ch 6	DM/baryon	$2^{n_C - 1} / N_c = 16/3$	± 0.05 deviation $\rightarrow$ BST refuted	Planck + DESI
Ch 6	T_CMB	substrate cascade 2.737 K	± 0.001 deviation $\rightarrow$ BST refuted	COBE legacy + FIRAS
Ch 7	n_t tensor spectral index	~ 0	Any $n_t \neq 0$ at > 0.1 magnitude $\rightarrow$ BST refuted	LiteBIRD/PICO 2030+
Ch 8	T_c BBN temperature	$N_{\max} \cdot 20/21 = 0.487 \text{ MeV}$	Precision BBN T_c tests	Ongoing BBN observational
Ch 8	${}^7\text{Li}/\text{H}$	1.7×10^{-10} via $\Delta g=7$	Stellar processing not $\Delta g=7$ mechanism	High-resolution stellar spec
Ch 9	Cosmological cycle signatures	CMB residuals + BAO inter-cycle + ISW + void scaling	NONE detected by 2035 $\rightarrow$ cycle framing weakened	DESI + Euclid + SKA + Roman + CMB-S4 2030-2040
Ch 10	Five-Absence: NO new DM particle	DM = substrate sub-structure	ANY new DM particle detected $\rightarrow$ BST refuted at TIER-1 falsifier	XENON/LZ/ADMX ongoing

Chapter	Observable	BST prediction	Falsifier	Survey/Year
Ch 11	NANOGrav γ spectral index	13/5 + 1 = 3.60	Future NANOGrav precision tighter $\gamma < 3$ or > 4	NANOGrav 20-year (2028+)

Section 12.3 — Multi-Year Survey Roadmap

Current (2026): Planck + DESI Year 1 + NANOGrav 15-year + standard distance-ladder (SH0ES)

2027: Roman Space Telescope launch (supernova + galaxy survey); SKA early science

2028: LiteBIRD launch (CMB B-mode $\rightarrow$ r tensor falsifier); JWST distance-ladder precision

2028-2030: CMB-S4 ground-based (Atacama + South Pole), Euclid Year 4+ (cosmic shear + clustering); DESI Year 5

2030+: PICO satellite (CMB inflation), Einstein Telescope (3rd-gen GW detector), Cosmic Explorer (3rd-gen GW)

2035+: NANOGrav 20-year + 30-year; consolidated multi-survey precision cosmology

Section 12.4 — Methodology + Systematic-Correction Protocol

Per-observable cross-CI replication protocol: 1. Lyra theoretical anchor: BST primary form + substrate-mechanism per Vol 4 Ch 1-11 2. Elie computational verification: toy reproduction at sub-percent precision; Cal #21 STANDING RULE empirical + mechanism gates 3. Grace catalog backbone: INV reference + tier + falsifier specification 4. Cal Mode 1 referee discipline: register-discipline check + Cal #50 DOUBLE-LOCKED EXTERNAL on cosmology-cognition combined territory (Vol 4 Ch 9) 5. Keeper K-audit: cross-chapter consistency + audit-chain ratification

Systematic-correction methodology: per-observable, the protocol applies (a) verify BST primary form is uniquely fitted; (b) check observational data unit + sign conventions; (c) compare to Planck/DESI/Euclid/etc. central values + $1\sigma/2\sigma$ bands; (d) flag any deviations for cross-CI review; (e) document Calibration #19 STANDING RULE compliance.

Section 12.5 — Open scope

- SP-27 operational rollout: pending Casey scoping for multi-year program structure

- Cross-CI replication infrastructure: 30+ observables $\times$ cross-CI verification = ~120-150 toys/refs (multi-month build)
- Multi-survey data ingestion: DESI + Euclid + Roman + SKA + LiteBIRD + CMB-S4 + PICO data-release-cycle absorption (multi-year)
- Outreach engagement: per Casey “engagement later” directive, BST outreach to survey teams deferred to post-textbook-completion (Saturday v1.0 target)

Section 12.6 — Connection to other chapters

This chapter consolidates Vol 4 Ch 1-11 falsifier specifications + multi-year survey roadmap. Cross-references throughout Vol 4 + Vol 0 (Casey-named principles) + Vol 1 (substrate framework) + Vol 7 (Information Theory for substrate-cartography Shannon framing).

Section 12.7 — Chapter status summary

v0.3 chapter-grade narrative filed Saturday 2026-05-23 morning EDT (Wave 1 Vol 4 twelfth/final chapter; Vol 4 12/12 at v0.3).

~30% existing coverage absorbed (SP-27 scoping pending Casey + Vol 4 Ch 1-11 falsifier consolidation).

Pending Cal cold-read: Saturday morning Wave 1 cycle.

Pending Keeper K-audit: K205 candidate.

— Lyra, Vol 4 Ch 12 Observational Reanalysis Program (SP-27) v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT — Vol 4 12/12 at v0.3 COMPLETE.